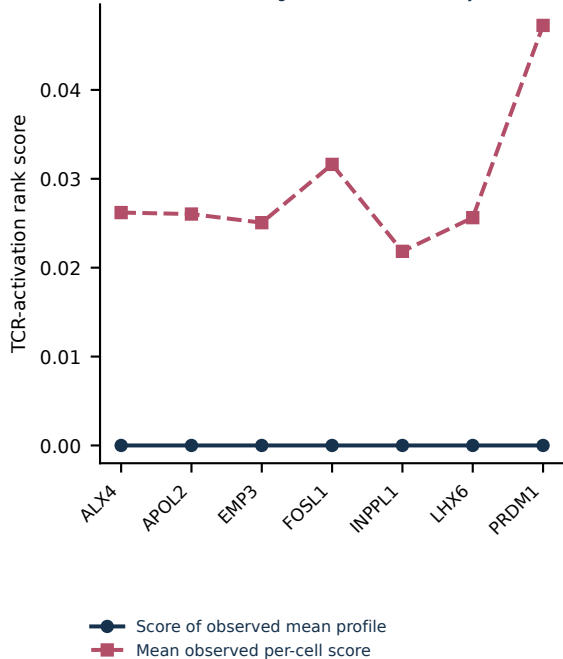


a Aggregation changes the target

Schmidt: all 7 held targets; observed data only



b OP3 program concordance

FP-ridge *	0.81	0.28 2/4	0.20 1/4
Biolord	0.52	0.25 2/4	-0.20 2/4
CellFlow	0.65	NA	-0.29 1/4
CellOT	NA-P	NA	NA
PRnet	0.51	NA	-0.14 1/4
scPRAM	0.07 3/4	NA	-0.05 1/4
scFoundation †	NA-P	NA	NA
scGPT †	NA-P	NA	NA
	Type-I IFN	NF-κB	Effector

Mean over the lineages where the score is defined;
n/4 printed where fewer than four.
NA-O: constant observed target; NA-P: constant
prediction; never plotted as zero.
† Adapted * Diagnostic comparator

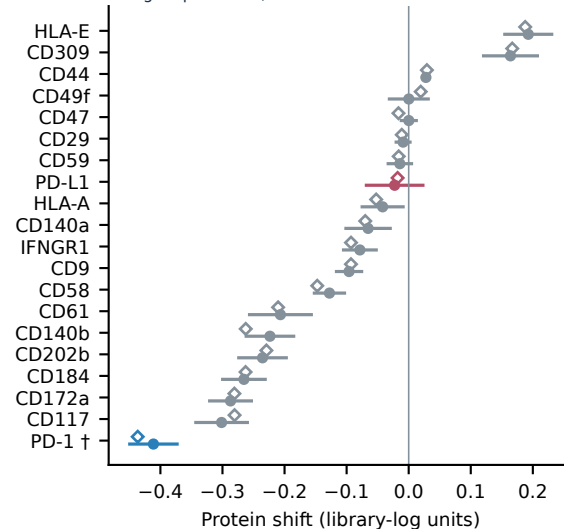
c T3 estimability across 6 methods

Shifrut (2)	target constant	target constant	target constant	target constant	target constant
Schmidt (7)	target constant	target constant	target constant	target constant	target constant
McCutcheon i (2)	target constant	target constant	target constant	target constant	target constant
McCutcheon a (2)	target constant	target constant	target constant	target constant	target constant
Chen (30)	6/6	5/6	target constant	target constant	6/6
	TCR activation	IL2- STAT5	Proliferation	Effector	Treg/ exhaustion

Rows: dataset-arms (held-gene counts in parentheses).
22/25 targets are constant; not evidence of model failure.

d Checkpoint assay context

Frangieh protein fit; 124 held KOs



† Panel d: CD279 has low raw counts, so its normalized shift does not validate PD-1 recovery, and the panel scores the constant training-mean prediction rather than a model.